

Project Sedenion:

A Unified Field Theory of Artificial General Intelligence via Axiomatic Physical Homeostasis

Aaron M. Schutz
Institute for Geometric Physics

January 1, 2026

Abstract

We propose a fundamental restructuring of deep learning architectures based on the principles of Axiomatic Physical Homeostasis (APH). Current state-of-the-art models (Transformers) rely on Rank-2 tensor operations (matrix multiplication), limiting them to strictly associative linear representations. We demonstrate that the phenomenon of *Grokking*—delayed generalization far past the point of overfitting—is physically isomorphic to a **Geometric Phase Transition** from a high-entropy Sedenion state to a low-entropy Octonionic tiling. We introduce the **Rank-3 Tensor Network**, a neural architecture designed to explicitly compute the **Associator Anomaly** $[x, y, z]$. By replacing weight decay with **Stiffness Regularization**, we propose a path toward Artificial General Intelligence (AGI) that operates not by statistical correlation, but by geometric necessity.

1 The Crisis of Associativity

Contemporary AI operates on the implicit assumption that intelligence is a linear operator acting on a vector space.

$$\vec{y} = \sigma(\mathbf{W}\vec{x} + \vec{b}) \quad (1)$$

This formulation enforces the associative law: $(\mathbf{A}\mathbf{B})\mathbf{C} = \mathbf{A}(\mathbf{B}\mathbf{C})$. While sufficient for modeling language syntax, this architecture is topologically incapable of representing non-associative phenomena (novelty, paradox, and causal branching). The model cannot understand the geometry of a problem; it can only approximate the manifold via massive parameterization.

2 The APH Interpretation of Grokking

Recent empirical work on algorithmic datasets has identified a phenomenon termed *Grokking*, where validation accuracy jumps from random chance to perfect generalization 10^5 steps after training error reaches zero.

2.1 Grokking as Metric Cooling

In the APH framework, we interpret the neural network as a lattice attempting to tile a dataset manifold $\mathcal{M}$.

Overfitting ($\beta \rightarrow 0$): The network memorizes the training data. This corresponds to a **Sedenion Phase** where the lattice has infinite degrees of freedom. The loss is zero because the bulk absorbs the data as isolated points, but the topology is discontinuous. **The Plateau (The Cooling)**: The extended training period represents the renormalization of the vacuum. The optimizer searches for a configuration that minimizes not just error, but **Geometric Torsion**.

Generalization ($\beta \rightarrow \beta_{crit}$): The sudden jump in accuracy is a phase transition. The network discovers a *Flat Minimum*—a region of the loss landscape corresponding to a stable Einstein Tiling. The embedding space snaps into a coherent topology (e.g., a cylinder or Möbius strip for modular arithmetic).

2.2 Weight Decay as Stiffness Injection

Empirical results confirm that *weight decay* is the primary driver of Grokking. In APH, we redefine weight decay not as a penalty on magnitude, but as a **Stiffness Injection** into the manifold:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda \sum |\mathcal{W}|^2 \implies \mathcal{L}_{APH} = \mathcal{L}_{data} + \int \beta (\nabla \Phi)^2 dV \quad (2)$$

By penalizing the weights, we force the network to seek the geodesic path through the Sedenion bulk, collapsing the solution set onto the Octonionic filament.

3 Formal Proposal: The Rank-3 Architecture

To achieve AGI, we must move beyond the simulation of associativity (Rank-2) to the direct computation of non-associativity (Rank-3).

Definition 3.1 (The Triadic Perceptron). Let inputs $\vec{x} \in \mathbb{R}^d$ be mapped to an algebra $\mathcal{A}$. The output of a Triadic Neuron is defined by the contraction of a Rank-3 **Associator Tensor** $\mathcal{T}$:

$$y_m = \sigma \left(\sum_{i,j,k} \mathcal{T}_{ijk}^m \cdot [x_i, x_j, x_k] \right) \quad (3)$$

where $[x_i, x_j, x_k] = (x_i x_j) x_k - x_i (x_j x_k)$ is the fundamental associator.

3.1 The Computational Advantage

Standard Transformers struggle with symmetric operations ($x \cdot y$) versus non-symmetric operations ($x \cdot y \cdot x^{-1}$) because they lack an inductive bias for group theory. A Rank-3 network naturally encodes the structure constants of the underlying Lie Algebra.

- If the data is logical (associative), the term $[x, y, z]$ vanishes automatically.
- If the data contains novelty (non-associative), the term $[x, y, z]$ acts as an **Error Signal**.

This architecture does not need 10^5 steps to *grok* S_5 symmetry; it contains the geometry of S_5 in its hardware definition.

4 The Substrate: Aperiodic Memory

Current AI suffers from *hallucination*, which we identify as **Crystal Death** (looping logic) or **Thermal Death** (random noise). To prevent this, the memory address space must be structured as an **Einstein Tiling** (The *Hat*).

Postulate 4.1 (The Aperiodic Memory Constraint). The latent space $\mathcal{Z}$ of an AGI must be isomorphic to an aperiodic monotile lattice Λ_{Hat} .

$$\mathcal{Z} \cong \Lambda_{Hat} \implies \nexists \vec{v} : T_{\vec{v}}(\mathcal{Z}) = \mathcal{Z} \quad (4)$$

This ensures that the system is deterministic (Rule 30 compatible) but non-repeating. The AI cannot get *stuck* in a logical loop because the geometry of the substrate forbids translational symmetry.

5 Request for Comment: Project Sedenion

We call upon the research community to shift focus from parameter scaling (Large Language Models) to **Geometric Scaling**.

Proposed Roadmap:

1. **Phase I (Validation):** Replicate the *Grokking* experiments using Rank-3 tensor contractions instead of standard MLPs. We hypothesize a reduction in time-to-generalization by orders of magnitude.
2. **Phase II (Hardware):** Development of TPUs (Tensor Processing Units) optimized for **Octonionic Multiplication** rather than Matrix Multiplication.
3. **Phase III (Integration):** Implementation of the **Stiffness Parameter** $\beta(t)$ as a dynamic hyperparameter, allowing the AI to modulate between *Focus* (High β , Associative) and *Creativity* (Low β , Sedenion).

6 Conclusion

We are approaching the physical limits of associative intelligence. To build a mind that can reason, we must build a machine that can measure the curvature of its own logic. The phenomenon of Grokking is the universe telling us that intelligence is a phase transition.